TECHNICAL ENGLISH • MODULE 1 • SESSION 6

The Corrective Action Plan & CAP Review

EASA Part-ADR.OR.D.005 Statutory Framework, Spoken Authority Negotiation & Session 7 Flipped Prep

📌 Lesson Summary & Operational Context

Closing an audit finding or incident investigation is not merely an administrative exercise—it is the statutory mechanism that prevents repeat catastrophic safety occurrences on the airfield. Under Regulation (EU) No 139/2014 Part-ADR.OR.D.005, when civil aviation authorities identify safety deficiencies, the aerodrome operator must formally demonstrate root-cause remediation, establish immediate interim risk mitigations (such as NOTAMs or operational restrictions), and agree on auditable verification milestones acceptable to the competent authority. In this capstone session, GSEA students examine an authentic EASA CAP dossier covering runway rubber hydro-blasting and jet blast deflector repairs, master formulaic commitment language (*we formally undertake to*, *subject to authority concurrence*, *in the interim*), simulate high-stakes regulatory negotiations with state inspectors, and dedicate extended workshop time to planning and structuring their 30-minute flipped peer-teaching lessons for Session 7.

Target Level: CEFR B1–B2 • **Track:** Technical English (AT • LV125E) • **Time:** 2 Hours (Teacher-Paced)

🎯 4 Core Learning Objectives for Session 6

1. Audit Finding Taxonomy & Root Cause: Distinguish Level 1 findings (immediate airside hazards, 24–48h NOTAM/restrictions) from Level 2 findings (systemic non-compliance, 30-day CAP) citing Part-ADR.OR.D.005.

2. CAP Drafting & Authority Benchmarks: Formulate SMART corrective actions, establish interim operational mitigations, and identify auditable physical verification criteria.

3. Spoken Regulatory Negotiation: Roleplay compliance meetings using formulaic spoken commitment phrases (*undertake to*, *in the interim*, *subject to concurrence*, *respectfully submit*).

4. Flipped Lesson Design Studio: Collaborate in pods of 3 to 4 with dedicated studio time to structure a 30-minute interactive peer lesson for Session 7 across authentic aerodrome domains.

📄 Worksheet Contents & Activity Index

Part / Stage	Activity Title	Focus & Interactive Learning Output	Page
Part 1	Audit Finding Taxonomy & Diagnostic	Statutory framework, Level 1 vs Level 2 & entry spot-check (4 questions).	Page 2
Part 2	Physical Remediation Gallery	3 photo exhibits: Fuel farm bunding, runway rubber hydro-blasting, blast deflector.	Page 3
Part 3	EASA Part-ADR.OR.D.005 CAP Dossier	Finding 041 Runway Friction, Finding 042 Blast Fence, Root-Cause Matrix & Sign-Off.	Page 4
Part 4	Language Clinic: Regulatory Commitment	5 formulaic negotiation phrases, reference table & 5 gap-fill practice drills.	Page 5
Part 5	Operator vs Inspector Spoken Negotiation	Roleplay simulation, Inspector peer checklist & 1-minute oral coaching protocol.	Page 6
Part 6 & 7	Safety Check, Rules to Reality & Studio	4-step decision table, 60-second wrap-up & Session 7 lesson design notes.	Page 7

PART 1 Activity 1: Aerodrome Audit Finding Notice: The Path to Compliance

Student Instructions & Action Steps:

1. **Step 1: Regulatory Briefing:** Read the EASA Part-ADR.OR.D.005 statutory framework below.
2. **Step 2: Complete the Diagnostic Spot-Check:** Answer the 4 core regulatory questions in pairs.
3. **Step 3: Pair Discussion:** Discuss why civil aviation authorities insist on auditable interim risk mitigations.

📋 Regulatory Briefing: EASA Part-ADR.OR.D.005 Framework

Under Regulation (EU) No 139/2014 Part-ADR.OR.D.005, receiving an audit finding notice triggers strict legal deadlines. Aerodrome operators cannot merely repair surface symptoms. They must isolate the underlying organizational root cause, determine whether flight safety is immediately hazarded (requiring provisional mitigations within 24 to 48 hours), and submit a formal Corrective Action Plan (CAP) detailing auditable engineering milestones acceptable to the competent civil aviation authority.

🔍 Opening Diagnostic: Audit Finding & CAP Principles

ENTRY SPOT-CHECK

Question 01: Level 1 vs Level 2 Audit Findings

Under Part-ADR.OR.D.005, what distinguishes a Level 1 finding from a Level 2 finding?

✓ Model Answer: Level 1 poses an immediate hazard requiring action in 24–48h; Level 2 is systemic non-compliance requiring a 30-day CAP.

Question 02: Interim Risk Mitigations

What is an 'interim risk mitigation' and why is it mandatory before permanent works begin?

✓ Model Answer: A provisional operational safeguard (e.g., NOTAM, speed limits) to contain risk while permanent engineering works are executed.

Question 03: CFME Friction Threshold

What friction coefficient threshold triggers a mandatory Level 1 finding for runway rubber build-up?

✓ Model Answer: When measured friction decays below the minimum planning level (mu < 0.34 under EASA CS-ADR-DSN.B.015).

Question 04: Authority Concurrence

What does 'authority concurrence' mean in the EASA CAP approval process?

✓ Model Answer: Formal written acceptance by the national authority (DGAC/EASA) confirming that proposed milestones meet regulatory standards.

🗣️ Pair Discussion: Negotiating with Aviation Regulators

Point A • Realistic Milestones: Why do civil aviation inspectors reject CAP submissions with vague deadlines like *"as soon as possible"*? (*Talking point: EASA standards require auditable target dates tied to physical verification*).

Point B • Commercial Realities: How can an aerodrome manager convince an inspector to grant a 30-day timeline without compromising safety? (*Talking point: By demonstrating that interim NOTAMs and speed restrictions fully mitigate operational risk*).

PART 2 Activity 2: Physical Remediation Gallery • Audit Finding Evidence

Student Instructions: Inspect the three airside engineering exhibits below. Examine the defect data, identify the operational risk, and formulate the immediate interim mitigation in spoken English.

Exhibit A: Fuel Bund — LEVEL 2

Finding 2026-ADR-038: LFBO Fuel Depot Tank Bund 03. Civil engineers inspecting sealant depth and concrete bund walls following joint degradation.

Operational Risk: Joint sealant degradation risks kerosene migration into stormwater drainage during rainfall.

Interim Mitigation: Daily visual inspection protocol and portable absorbent spill-booms deployed around Tank 03 pending elastomeric re-sealing.

Exhibit B: Hydro-Blasting — LEVEL 1

Finding 2026-ADR-041: Runway 14L Touchdown Zone (0–600m). Specialized UHP hydro-blasting truck (2,200 bar) stripping vulcanized tire rubber deposits.

Operational Risk: Friction decayed to mu = 0.31 (below 0.34 threshold), creating dynamic hydroplaning risk during wet landings.

Interim Mitigation: Permanent NOTAM prohibiting wet tailwind landings; airlines apply +15% landing distance buffer until mu >= 0.52.

Exhibit C: Blast Deflector — LEVEL 2

Finding 2026-ADR-042: Holding Point P1 Perimeter. Maintenance crew replacing corroded curved blast panels and installing high-tension nylon FOD catch mesh.

Operational Risk: Corroded anchor bolts risk structural panel detachment under 120-knot widebody breakaway engine thrust.

Interim Mitigation: Adjacent service road checkpoint Delta closed; ATC enforces idle breakaway thrust taxi restrictions at Holding Point P1.

🔍 Formulating SMART Corrective Milestones

- **Specific:** Define the engineering intervention clearly (*ultra-high-pressure hydro-blasting at 2,200 bar*, not *runway cleaning*).
- **Measurable:** Link completion to certified criteria (*friction traces confirming mu >= 0.52, 45 kN pull-out torque*).
- **Achievable & Realistic:** Schedule realistic nighttime closures without disrupting scheduled commercial traffic.
- **Time-Bound:** State the exact target date for on-site authority verification (*28 October 2024*).

PART 3 Activity 3: The EASA Corrective Action Plan (CAP) Dossier

Statutory Framework: EASA Part-ADR.OR.D.005 mandates that operators submit a detailed CAP isolating the root cause, defining auditable interim mitigations, and securing authority concurrence before closing findings.

Finding 041: Runway Friction Decay

LEVEL 1 • CS-ADR-DSN.B.015

Survey Data: Continuous Friction Measuring Equipment (CFME) recorded average mu = 0.31 at 65 km/h with 1.0mm water film across the first 600m. Grooves 85% rubber-filled.

Interim Mitigation: NOTAM issued: *"RWY 14L FIRST 600M BRAKING ACTION POOR WHEN WET. TAILWIND LANDINGS PROHIBITED EXCEEDING 5 KT."* Airlines apply +15% dispatch buffer.

Definitive Remediation: 4 nights of UHP hydro-blasting (2,200 bar) stripping 28,000 m² of rubber; 3 CFME calibration runs at 65 & 95 km/h. **Target: 28 Oct 2024.**

Finding 042: Blast Deflector Failure

LEVEL 2 • CS-ADR-DSN.D.260

Defect Data: Holding Point P1 blast deflector shows severe anchor bolt corrosion. Three corrugated panels unseated; top FOD netting torn over 40-meter stretch.

Interim Mitigation: Service road Delta closed during widebody departures; ATC instructs all aircraft at P1 to maintain idle thrust until aligned with centerline.

Definitive Remediation: Bolting Grade 316 stainless anchors rated for 45 kN pull-out torque, new corrugated curved panels, and high-tension nylon FOD mesh. **Target: 30 Nov 2024.**

Root-Cause Remediation Matrix

Non-Conformity	Identified Root Cause	Immediate Mitigation	Permanent Engineering Fix	Deadline
Runway Friction Restoration	Annual fixed maintenance schedule failed to account for 30% surge in summer widebody traffic.	**Issue NOTAM; enforce landing performance dispatch buffer.**	Switch to dynamic bi-monthly CFME monitoring; execute UHP hydro-blasting.	**28 Oct 2024**
Blast Fence Anchoring	Original 2012 specification lacked marine-grade epoxy protection against de-icing chemical runoff.	**Close adjacent service road; enforce idle thrust taxiing.**	Bolt 45 kN pull-out rated stainless anchors; install continuous FOD mesh.	**30 Nov 2024**
Fuel Farm Containment	Elastomeric joint compound reached end of operational lifespan without scheduled re-caulking.	**Deploy portable absorbent booms around Tank 03; daily visual checks.**	Hydro-cleaning and application of aviation-grade chemical-resistant polyurethane.	**15 Nov 2024**

DGAC / EASA Oversight Directorate Sign-Off Statement

PROVISIONAL CONCURRENCE

"Level 1 closure is conditional upon physical verification of CFME mu >= 0.52 across all three runs. Level 2 closure requires submission of certified pull-out torque test certificates (45 kN/anchor) and photographic proof of completed FOD mesh installation. Provisional concurrence granted." — Lead Aviation Safety Inspector #FR.DSAC.2024-04

PART 4 Activity 4: Commitment Phrases for Regulatory Negotiation

Language Focus: When negotiating CAP milestones with civil aviation inspectors, aerodrome managers use precise formulaic commitment markers to balance legal compliance with operational reality.

📖 5 Formulaic Regulatory Commitment Phrases

Commitment Phrase	Negotiation Function	Authentic Airport Example	Grammar Pattern
`we formally undertake to`	Expresses a legally binding operational commitment to complete works	*"We formally undertake to complete hydro-blasting within 14 calendar days."*	undertake to + [base verb] within [time]
`in the interim, the operator shall`	Establishes an immediate provisional safeguard active before permanent works	*"In the interim, the aerodrome operator shall enforce a mandatory NOTAM."*	In the interim, [subject] shall + [verb]
`subject to authority concurrence`	Conditions an operational schedule upon formal civil aviation approval	*"Subject to your concurrence, our engineering team will submit weekly photos."*	Subject to [party's] concurrence, we will...
`conditional upon physical verification`	Specifies that finding closure depends on auditable test proof	*"Finding closure is conditional upon physical verification of CFME traces."*	conditional upon [noun/test result]
`we respectfully submit that`	Introduces a polite, diplomatic pushback against excessive restrictions	*"We respectfully submit that closing the entire runway is disproportionate."*	We respectfully submit that [clause]

✍️ 5 Practice Drills: Select the Correct Regulatory Phrase

1. The aerodrome operator _____ complete ultra-high-pressure rubber removal across the first 600 meters within 14 nights.
Options: [formally undertakes to | respectfully submits that | in the interim] • **Answer: formally undertakes to**

2. _____, air traffic control shall restrict all aircraft at Holding Point P1 to idle breakaway thrust.
Options: [Subject to concurrence | In the interim | Conditional upon] • **Answer: In the interim**

3. The revised 45-day blast fence reconstruction schedule is valid only _____ from the state oversight directorate.
Options: [subject to authority concurrence | we respectfully submit that | in the interim] • **Answer: subject to authority concurrence**

4. Formal Level 1 finding closure remains _____ on-site CFME calibration runs confirming mu >= 0.52.
Options: [conditional upon physical verification of | we formally undertake to | in the interim] • **Answer: conditional upon physical verification of**

5. _____ imposing a 30-day runway closure is commercially unviable given our verified nighttime hydro-blasting capacity.
Options: [In the interim | We respectfully submit that | Subject to concurrence] • **Answer: We respectfully submit that**

💡 Negotiation Tip: Balancing Assertiveness and Regulatory Respect

Avoid Confrontation: Never say: *"Your deadline is impossible and will cost us too much money."*
→ Use Professional Diplomacy: *"We respectfully submit that an immediate full closure is disproportionate given our interim NOTAM buffer, and we formally undertake to complete nightly hydro-blasting within fourteen calendar days."*

PART 5 Activity 5: Aerodrome Operator vs Civil Aviation Inspector Negotiation

Simulation Instructions: Work in pairs. Partner A acts as the Aerodrome Safety & Compliance Manager defending the proposed CAP schedule; Partner B acts as the Civil Aviation Inspector enforcing safety standards under Part-ADR.OR.D.005.

Role A: Aerodrome Safety Manager — OPERATOR

Goal: Present proposed CAP milestones, justify why blast fence procurement requires 30 days due to specialized alloy delivery, and prove that interim NOTAMs fully contain flight safety risks.

Target Spoken Phrases:
- *"We formally undertake to complete hydro-blasting within 14 calendar days."*
- *"In the interim, the operator has published a NOTAM prohibiting tailwind landings."*
- *"Subject to your concurrence, our team will submit weekly progress photos."*
- *"We respectfully submit that closing the runway is commercially disproportionate."*

Role B: Civil Aviation Inspector — DGAC / EASA

Goal: Scrutinize the CAP timeline, demand verifiable proof, verify that Level 1 risks are mitigated within 24 hours, and grant conditional concurrence only with auditable progress checks.

Target Inspector Phrases:
- *"The authority acknowledges receipt, but 30 days for Level 1 friction is excessive."*
- *"We require physical CFME traces showing mu >= 0.52 before the NOTAM can be cancelled."*
- *"Our concurrence is conditional upon certified torque pull-out test reports for all 40 bolts."*
- *"What specific contingency is in place if the hydro-blasting vehicle suffers downtime?"*

Civil Aviation Inspector Peer Evaluator Checklist — PEER EVALUATION

Partner B evaluates Partner A's compliance negotiation against these 4 statutory criteria before giving 1 minute of oral coaching:

- ☐ 1. Finding Classification: Accurately cited Level 1 immediate vs Level 2 30-day CAP?
- ☐ 2. Commitment Phrases: Accurately used *undertake to*, *in the interim*, *subject to*?
- ☐ 3. Realistic Milestones: Proposed SMART auditable dates & enforceable NOTAM buffers?
- ☐ 4. Assertive Diplomacy: Maintained professional composure without aggressive pushback?

Inspector (Evaluator): "Your interim NOTAM strategy for Runway 14L was very thorough. However, when proposing the 30-day blast fence timeline, remember to explicitly use formulaic commitment phrases like: *we formally undertake to submit weekly progress photos subject to authority concurrence*."

Safety Manager (Operator): "Thank you for that observation, Inspector. We formally undertake to provide certified torque pull-out test reports every seven days, conditional upon your on-site verification."

Structure of a Professional Regulatory Negotiation

- **Opening Acknowledgment:** Formally acknowledge receipt of the finding reference number and classification.
- **Immediate Containment:** State the interim operational restriction already active (NOTAM, speed caps).
- **Definitive Engineering Fix:** Explain the root-cause remediation with specialized contractor equipment.
- **Binding Commitment:** Conclude with a formulaic deadline commitment subject to authority verification.

PART 6 Activity 5: Safety Check: What is the Airfield Risk?

Simple Safety Check: Review the 4-step decision table below to evaluate audit findings, corrective action plans, and regulatory enforcement decisions.

Simple Safety Question	Agreed Investigation Board Finding & Decision
1. What is the systemic problem?	Runway 14L touchdown zone has decayed to mu = 0.31 (below 0.34 safety threshold) due to rubber build-up; Holding Point P1 blast deflector has corroded base bolts risking fence detachment under 120-knot jet thrust.
2. What could happen if uncorrected?	Decayed friction risks hydroplaning runway excursions during wet landings; detached blast fence panels risk launching high-speed metallic FOD across Taxiway Papa and adjacent service roadways.
3. How do we fix it?	Enforce immediate NOTAMs (no tailwind landings, idle thrust taxiing); execute 4 nights of UHP hydro-blasting (28,000 m²) and install Grade 316 stainless anchors rated for 45 kN pull-out torque.
4. Directorate Decision	Grant provisional EASA/DGAC concurrence: Level 1 closure is conditional upon CFME mu >= 0.52 verification; Level 2 closure is conditional upon certified anchor pull-out test certificates.

⚡ Rules to Reality: The 60-Second Compliance Pitch UNSCRIPTED SPOKEN WRAP-UP

Without looking at your notes, turn to your partner and explain in 60 seconds: *"Why can't an aerodrome operator simply promise to fix a finding next year?"* You must use at least 2 commitment phrases (*undertake to*, *in the interim*, *subject to concurrence*) and distinguish a Level 1 hazard from a Level 2 non-compliance.

PART 7 Activity 6: Session 7 Flipped Classroom Planning Studio

1. Chosen Domain, Lesson Title & Opening Hook (~5 Mins):

2. Core Communicative Activity & Step-by-Step Instructions (~18 Mins):

3. Team Member Role Allocation & Classroom Delivery Strategy: